

Response Letter with Annotations

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Annotations

1. Purposes

- a. Motivation (one-sentence summary)
 - The study is motivated to understand whether the real-world mix of media messages about tobacco actually shapes young people's intentions to smoke, beyond the effects of formal anti-tobacco campaigns.
- b. Specific Rationale (several bullet points)
 - Smoking remains a major public health problem.
 1. Understanding how the accumulation of everyday media content discouragement or encouragement of youth smoking's behavior has public health significance.
 - Traditional media-effects research often isolates messages based on class (i.e. source or theme) in experiments, which often does not reflect the messy real communication environment where people see many, overlapping messages over time.
 - Many observational media studies rely on self-reported exposure, raising endogeneity and causal order concerns
 1. This study aims to build stronger causal inference in using exogenous media measures for media exposure which are not contingent on individual participant reports
- c. Research Questions and/or Hypotheses
 - Identify IV, DV, mediator, moderator
 1. IV: Main predictors are exogenous measures of the public communication environment (PCE):
 - a. Anti-tobacco media coverage index: aggregated past 7-day anti-tobacco content from AP, broadcast news, newspapers, websites, and Twitter, standardized via principal components
 - b. Pro tobacco media coverage index: parallel measure for pro tobacco content
 - c. Anti and pro tobacco YouTube views: logged counts of views of valenced tobacco videos in the prior 7 (or 28) days
 2. DV:
 - a. Intentions to smoke in next 6 months: combined measure where any intention to smoke (nonsmokers) or not definitely intending to quit (smokers) was assessed

- b. Anti-smoking belief scale: mean of multiple items on negative consequences of daily smoking
 - 3. Moderators:
 - a. Established smoking status (current established smoker vs. not). It significantly moderates the effect of anti-tobacco coverage on intentions
- Hypotheses
 - 1. (H1) exposure to tobacco information in the PCE opposed to use (anti) and supportive of use (pro) will affect respondents' intentions to use combustible cigarettes (with anti-tobacco content reducing intentions and pro-tobacco content increasing intentions).
 - 2. (H2) exposure to such valenced tobacco content will also affect endorsement of beliefs about combustible cigarettes for both smokers and nonsmokers (with anti-tobacco content increasing endorsement of anti-tobacco beliefs and pro-tobacco content reducing endorsement of those beliefs)
- Research Question
 - 1. (RQ1) We ask whether hypothesized effects are moderated by respondent smoking history: will current smokers and current nonsmokers be differentially affected by valenced tobacco content, so that the decision to initiate and the decision not to quit (both of which represent intentions to smoke in the next 6 months) will show different influence of media content?

2. Method

a. Research Design

- Observational, time-series, rolling cross-sectional study. National rolling cross-sectional phone surveys were conducted daily across 1,147 days between 2014–2017. Each respondent was described the PCE coverage around tobacco during phone surveys.
- PCE coverage of long-form media like newspapers, AP, broadcast news, and web blogs texts were analyzed between May 2014 to June 2017. Amazon Mechanical Turk coders completed 2,400 codes for valance coding to train the machine learning.
 - 1. For Twitter content was simply counted as pro or anti-tobacco. And on YouTube view data was logged and labeled as anti or pro within the first 30 days of publication. For YouTube content, available text was evaluated, but not visual content.

b. Sample

- The sample of participants was a nationally representative sample of 13- to 25-year-olds (N=11,847)
 - 1. 38% of participants (n = 4,470) completed a second interview six months after their first interview.

c. Measures

- Media / PCE measures

1. Daily content from six sources: 50 newspapers, AP, 8 major broadcast news outlets, 100 youth popular websites, Twitter, and YouTube
 - a. For AP, broadcast, newspapers, web: daily sums of probabilities of being anti or pro smoking
 - b. Twitter: counts of anti or pro smoking tweets
 - c. YouTube: logged daily views of anti or pro smoking videos
 2. Automated supervised machine learning classified items as anti or pro tobacco
- Survey measures
 1. Established smoking status among participants
 - a. Have you ever tried smoking cigarettes, even one or two puffs?
 - b. Have you smoked at least 100 cigarettes, which is 5 packs, in your entire life?
 - c. During the past 30 days, on how many days did you smoke cigarettes?
 2. Intentions: Binary indicator of any intention to smoke in next 6 months (for nonsmokers) or not intending to quit (for smokers)
 - a. Established smokers were asked:
 - i. How likely is it that you will try to quit smoking completely in the next 6 months? By completely, I mean not smoking tobacco cigarettes at all. Would you say definitely will not, probably will not, probably will, or definitely will?"
 - b. Nonsmokers were asked:
 - i. How likely is it that you will smoke a tobacco cigarette, even one or two puffs, at any time in the next 6 months? Would you say definitely will not, probably will not, probably will, or definitely will?"
 3. Beliefs scale: 11 item anti-smoking belief index (e.g., cancer, addiction, looking uncool), 4-point Likert
 - a. The next set of questions is about tobacco cigarettes. I'll read a statement, then please tell me whether you strongly disagree, disagree, agree, or strongly agree with it. If I smoke every day, I will . . .
 - i. develop headaches
 - ii. develop sexual and/or fertility problems
 - iii. develop cancer
 - iv. get wrinkles
 - v. lose my teeth
 - vi. get yellow fingers
 - vii. become addicted to nicotine
 - viii. be controlled by smoking
 - ix. look uncool
 - x. [It will] be a turn off to other people
 - xi. feel relaxed
 - xii. enjoy life more

d. Analysis

- Logistic regression clustered by interview date, using exogenous media measures to reduce common-method bias and clarify causal order

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- Hypotheses
 1. H1: Partial support
 - a. Exposure to anti- tobacco information in the PCE *lowered* participants intentions to smoke.
 - b. Exposure to pro-tobacco information in the PCE had *no significant effect* on audiences intention to smoke.
 2. H2: Not supported
 - a. Exposure to these valenced (anti vs. pro) tobacco content *had no significant effect* on beliefs about cigarettes
- Research Question
 1. Yes, but only for anti-coverage. The negative effect of anti-tobacco coverage on intentions is stronger among current smokers than nonsmokers

b. Include Key Figures/Tables

Table 1 Content Sources and Validity Measurements

Source	Period of Measurement	Archive, Method for Locating Eligible Texts	Validity of Product Coding Texts	Method for Assessing Valence	Validity of Coding Valence
Broadcast news	18 May 2014 to 30 June 2017	Lexis-Nexis, 12 keywords	$P = .97, R = 0.98$	SML	$P_{pro} = 0.83; R_{pro} = 0.92$ $P_{anti} = 0.86; R_{anti} = 0.91$
Associated Press	18 May 2014–30 June 2017				
Newspapers	18 May 2014 to 30 June 2017				
Website blogs	18 May 2014 to 30 June 2017	Massachusetts Institute of Technology MediaCloud, 12 keywords			
Twitter	18 May 2014 to 30 June 2017	Gnip Twitter Historical Powertrack, 889 tags, keywords, rules	$P = .92, R = 0.95$	SML	$P_{pro} = 0.92; R_{pro} = 0.92$ $P_{anti} = 0.90; R_{anti} = 0.90$
YouTube	30 July 2014 to 30 June 2017	YouTube search application program interface, 220 keywords, rules	$P = .85, R = 0.87$	SML classification of videos	$P_{pro} = 0.96; R_{pro} = 0.93$ $P_{anti} = 0.60; R_{anti} = 0.93$

SML, supervised machine learning; P, Precision; R, Recall.

Table 2 Univariate Information for All Included Variables

Variable	Mean (SD)	Mean (SD)
Source—past seven days	Anti-tobacco texts	Pro-tobacco texts
Broadcast news	0.20 (0.22)	0.05 (0.10)
Associated Press (AP)	0.93(0.44)	0.05 (0.07)
Newspapers	3.62 (1.28)	0.57 (0.33)
Websites	5.31(1.94)	1.07 (0.56)
Twitter	12,948 (3,457)	21,336 (5,546)
YouTube views	79,773 (96,229)	72,317 (54,640)
Intention to smoke among non-smokers (% some intention)	22% (<i>n</i> = 10,369)	
Intention not to quit among smokers (% some intention not to quit)	80% (<i>n</i> = 1,422)	
Intention open to smoking among all	29% (<i>n</i> = 11,791)	
Established smokers (lifetime more than 100 and current)	12% (<i>n</i> = 11,847)	
Anti-tobacco belief scale (1–4) mean (<i>SD</i>)	3.00 (0.43) (<i>n</i> = 11,847)	

Note: Sources were summed for 7 days prior to interview date. The reported numbers represent the total pro- or anti-tobacco coverage in each source over that period. The means are estimated at the person-level. Survey measures reflect population weighting.

Table 3 Effects of Media Coverage on Intentions to Smoke, Main Effects, and Interaction

Predictor	Odds Ratios: Main Effects Model	Confidence Interval (CI)	Odds ratios: Model with Smoking Status Interaction	CI
Standardized time	0.92	0.83–1.02	0.92	0.83–1.02
Established smoking status	14.21	11.84–17.06	14.26	11.87–17.12
Anti-media coverage index	0.95	0.91–1.00	0.97	0.93–1.01
Pro-media coverage index	1.02	0.99–1.05	1.02	0.99–1.05
Anti-YouTube views	0.96	0.89–1.04	0.96	0.89–1.04
Pro-YouTube views	1.04	0.95–1.12	1.04	0.95–1.13
Established smoking status* Anti-media coverage index	—	—	0.88	0.77–1.00
Constant	0.28	0.26–0.29	0.28	0.26–0.29
<i>N</i>	11,343		11,343	

Note: Logistic regression, clustered by date, significant ($p < .05$) predictors bolded. The media coverage index is a standardized scale made up of Associated Press, Broadcast news, newspapers, websites, and Twitter. YouTube views are logged. Smoking status was only a significant moderator for the anti-media coverage index.

*It indicated that this is an interaction term – the product of Established smoking status and the Anti-media coverage index.

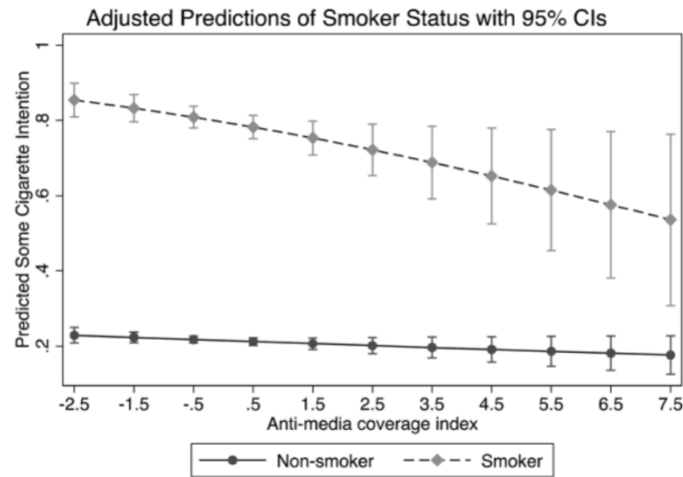


Figure 1 Adjusted predictions of smoker status with 95% CIs.

Table 4 Effects of Media Coverage on Anti-Smoking Beliefs, Main Effects

Predictor	Model with Main effects	
	B	Confidence interval
Standardized time	0.017	−0.004 to 0.038
Established smoker	−0.409	−0.444 to −0.373
Anti-media coverage index	−0.004	−0.012 to 0.005
Pro-media coverage index	0.002	−0.005 to 0.008
Anti-YouTube views	−0.014	−0.028 to 0.000
Pro-YouTube views	−0.002	−0.020 to 0.016
Constant	3.106	3.096 to 3.117
N	11,381	

Note: Ordinary least squares regression, clustered by date, significant ($p < .05$) predictors bolded. The media coverage index is a standardized scale made up of Associated Press, Broadcast news, Newspapers, Websites, and Twitter. YouTube views are logged. Smoking status was not a significant moderator of any media coverage variables.

c. Authors' Conclusions

- Routine anti-tobacco media coverage in the broader PCE does meaningfully reduce young people's intentions to smoke, especially among current smokers, even though the effects are small. Furthermore, tobacco coverage across valenced had no significant effect on audiences beliefs about smoking. The negative association between anti-tobacco coverage and intentions is larger for established smokers than for nonsmokers, suggesting smokers are particularly responsive to shifts in the PCE. Pro-tobacco coverage and YouTube measures do not show consistent significant effects, which can partially be attributed to low volumes of pro-content and very low

probabilities that any given youth actually sees the specific tobacco-related YouTube videos measured.

4. Discussion

a. Implications

- Findings in this work suggest that routine anti-tobacco content can affect smoking behaviour across a messy media environment, especially for established smokers.
 1. Even though effect sizes are small, anti-tobacco content should be seen as a complementing factor to smoking cessation campaigns as this content is more frequently presented to populations.
- Additionally, this work has important methodological implications. Given the difficulty for previous studies to build causal claims of media effects in accurate media environments, this study's methods can yield more causal inferences, especially in real-world contexts

b. Limitations

- Effect sizes are small; supervised machine-learning valence coding and incomplete coverage of all media introduces measurement error
 1. Content analysis of the six media source categories are not fully comprehensive of the PCE environment surrounding tobacco content
- Content variables capture opportunities for exposure, not what each respondent actually saw
- Smoking beliefs and intentions were already highly stable over time, making additional change harder to detect
- Causal assumptions in this study can only met if two things are true:
 1. Media content does not change due to individual intentions. Week-to-week variation in media attention to tobacco should reflect *external events* and not a direct response to population changes in *intentions* to smoke cigarettes
 2. Other variables do not affect media source coverage and individual intentions.

c. Future Directions

- Develop content coding for norms and self-efficacy pathways, not just outcome beliefs.
- Improve measurement of individual engagement with the PCE and test social network and institutional mechanisms more directly
- Apply the PCE approach to other health behaviors and risk topics

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Response Letter (~600 words)

- Your assessment of the study

Overall, I thought this study built really beautifully off of previous media effects research focusing on smoking behavior. I was particularly impressed with the rationale and explanation behind this study, as it justified the “so what” behind this work and why utilizing a computational method was so important to advance this field of study. Aligned with this, I thought the rationale behind their methodological choices to mitigate any endogeneity concerns was well-described. I was initially unaware of the terminology “endogeneity”, but I appreciated how authors here took some space to explain what this concept meant. Throughout this entire study, I easily understood what was happening and the reasoning behind each methodological choice. While it’s simple, I see this as one of the study’s largest strengths as this work could be comprehensible to audiences outside of research and potentially be utilized in campaign building and assessment.

Additionally, this work’s findings are fascinating and especially important to the field of smoking cessation and health communication. The finding that anti-tobacco content did decrease established smokers’ intentions to smoke is a nuanced result and indicates that viewing health content outside of a campaign context is mainly impactful to those currently engaged in particular behaviors. This is important on a health communication basis as this indicates how the accumulating media environment surrounding a health issue affects audiences’ intention to perform unhealthy behavior. Furthermore, it’s interesting to note on a social norms perspective that anti-tobacco media content had little effect on nonsmokers as a preventative measure. This may be due to the importance of self-relevance when it comes to health communication and how this content hooks in established smokers.

- Weaknesses

While this work has many strong suites, there are some weaknesses that can’t be ignored. To start, the evaluation of YouTube available text data while fitting for supervised machine learning practices, however, since researchers decided not to evaluate visual aspects here may have missed components of evaluation. For example, the title or description of a YouTube video could be coded as anti, but what’s inside the contents of the video are pro-smoking. Essentially, I see this as a potential weakness as this evaluation tactic may not be truly representative of the data.

Similar to what this study suggested in the limitations, the causal inferences of this work are limited due to several factors. As this study stated, the only way for causality to be true in this work is if (1) media focuses on tobacco are reflective of solely external events and not the general populations’ intentions on smoking; and (2) if other variables do not affect media source coverage and individual intentions. This second aspect is the main object of concern, because there are several aspects that can affect media coverage and individual’s intentions on smoking. For example, participants could’ve decreased their intention to smoke based on a variety of internal factors (i.e. social norms, change in environment, or reductions in stress) or varying media exposure beyond tobacco content (i.e. lung cancer campaigns or lifestyle influencers). This study makes great leaps from previous work that lacks endogeneity, however, this study like many others has a hard time crafting clear and definitive causal claims.